

Session 18 : AI in Everyday Life & Across Industries

Over the last four sessions, you've become fluent in the languages of AI perception. You know how AI reads text, sees images, hears voices, and combines all three. Now it's time to take those skills out of the lab and into the world. Today we're going to see where AI is present in your daily life and how it's reshaping entire industries.



Objectives

By the end of this session, you will be able to:

01

Spot AI

Identify **AI systems** in your daily life and across industries.

02

Apply the Framework

Use the **Sense → Think → Act** framework to analyze AI across major industries.

03

Use the Tools

Use **Gemini** for broad research and **NotebookLM** for fact-verification on an industry topic.

04

Human Judgment

Explain why **human judgment** remains essential even as AI becomes more powerful.

Recap

Domain	Framework	Tool You Used
NLP	Break → Understand → Generate	QuillBot, NotebookLM
Computer Vision	See → Process → Act	Teachable Machine
Voice AI	Hear → Process → Speak	ElevenLabs
Multimodal AI	Sense → Think → Act	Gemini, Veo 3, Pictory

Warm-Up: How many times did you interact with AI before today?

 **Weather app
forecast?**

AI prediction model

 **Face ID to unlock
phone?**

Computer Vision

 **Smart speaker news
briefing?**

ASR/STT + TTS

 **Scrolling social
feed?**

Recommendation
algorithm

 **Email spam filter?**

ML classification

 **Maps predicting commute
time?**

Regression model

 **Streaming service suggesting
a morning playlist?**

Recommendation algorithm



AI in Everyday Life

AI isn't just a chatbot or a robot. It's a layer of intelligence quietly woven into the apps and devices you use every single day - often without you even noticing. Before we explore entire industries, let's uncover the AI that's already in your pocket, your home, and your daily routine.

The Everyday AI Stack

AI isn't just a chatbot. It's a layer of intelligence woven into the apps and devices you already use.

Daily Action	AI Superpower	Domain
Taking a photo in Portrait Mode	Blurring background by estimating depth	Computer Vision
Autocorrect changing "teh" to "the"	Predicting the intended word	NLP (Language Model)
Searching photo library for "sunset"	Matching text query to image features	Multimodal AI
Voice typing a text message	ASR/STT → NLU	Voice AI
"You might also like..." suggestions	Finding patterns in similar users' behavior	Machine Learning

From Your Pocket to the Planet

The exact same AI technologies you've used for fun are deployed at massive scale:

NLP at Scale

The NLP that wrote your birthday poem (S7) scans **millions of medical records** to find rare disease patterns.

Computer Vision at Scale

The Computer Vision that recognized your hand gestures (S6) inspects **every single strawberry** on a conveyor belt for bruising.

Voice AI at Scale

The Voice AI that made a funny wizard voice (S16) powers **communication devices** for people who cannot speak.

 **Critical Reminder:** These systems are not alive. Face ID does not *know* you - it knows a mathematical pattern of your face is the most probable match. AI is a pattern-matching engine, not a thinking being.

Smart Home Thermostat

Let's break down how a familiar AI system works using the Sense→Think→Act model.



SENSE

Temperature sensor reads **62°F**
indoors at **6:45 AM**



THINK

AI model predicts: *"Based on past behavior, the homeowner wants it 70°F by 7:00 AM. It will take 12 minutes to warm up."*



ACT

Signals furnace to start heating at
6:48 AM

Mini Activity 1: Decode the Hidden AI

For each real-world scenario below, your job is to:

1 Identify the Primary AI Domain

Is it NLP, Computer Vision, Voice AI, or Multimodal?

2 Write the Sense → Think → Act Flow

Describe the full AI process in one clear, complete sentence.



Mini Activity 1: Decode the Hidden AI



Smart Fridge

Your refrigerator notices you're low on milk and automatically adds it to your shopping list.



Live Captions

While watching a live news broadcast, words appear at the bottom of the screen matching exactly what the reporter is saying.



Photo Search

You type "birthday cake" into your phone's photo app, and it instantly shows every picture you've ever taken with a birthday cake in it.



Noise-Canceling Earbuds

They block out the roaring airplane engine but let your friend's voice pass through clearly so you can have a conversation.

Mini Activity 1 - Answer Key

Scenario	AI Domain	Sense → Think → Act
Smart Fridge	Computer Vision	Sense: Camera captures milk level. Think: AI estimates volume and predicts depletion. Act: Sends "Add milk" to shopping list app.
Live Captions	Voice AI (ASR/STT)	Sense: Mic captures speech as sound waves. Think: ASR converts waves → phonemes → words in real time. Act: Displays transcribed text as captions.
Photo Search	Multimodal AI	Sense: Text query tokenized; image library encoded. Think: Joint embedding aligns text and image features. Act: Retrieves and displays matching photos.
Noise-Canceling Earbuds	Voice AI	Sense: External mic captures drone; internal mic captures voice. Think: AI generates anti-noise wave for low frequencies. Act: Speakers cancel engine sound while passing through voice.



AI Across Industries

We've seen AI in our pockets and our homes. Now let's zoom out. How are these same AI superpowers transforming entire industries, the systems that move people, save lives, educate minds, entertain millions, and power the global economy? Each industry below uses AI in uniquely powerful ways, but all follow the same fundamental patterns you've already mastered.



Healthcare



Finance



Education



Entertainment



Transportation



AI in Healthcare

Hospitals and clinics use AI to help doctors analyse medical images like X-rays, CT scans, and MRIs. These AI models are trained on thousands of labelled images - showing both healthy patients and those with diseases.

What AI Does: Analyzes medical images, predicts patient risk, accelerates drug discovery.

AI in Healthcare

Example

A radiologist uploads a chest X-ray. The AI model highlights a faint **3mm nodule** near the lung periphery - something easily missed by a tired human eye scanning hundreds of images. It's an early sign that requires follow-up.

Sense → Think → Act

- **Sense:** X-ray machine captures high-resolution image
- **Think:** CV model (trained on millions of labeled X-rays) detects small, round, high-density region
- **Act:** Flags the image "Priority Review" for the radiologist

- ☑ **Human-in-the-Loop:** The radiologist confirms the finding, orders additional tests, and decides treatment. AI is the scout, not the doctor.



AI in Finance

Banks and credit card companies use AI to protect your money from fraud. Every time you swipe your card, the AI analyzes dozens of details: the purchase amount, the location, the time of day, the type of store, and whether you've shopped there before. It compares this to your normal spending patterns - learned from your past transactions and flags anything that looks unusual.

What AI Does: Detects fraud, assesses credit risk, powers algorithmic trading.

AI in Finance

Example: Credit Card Fraud Detection

You're on vacation and buy a product in a foreign country. Seconds later, a second transaction attempts to buy a **\$2,000 laptop** online. The AI model flags the second transaction as suspicious.

Sense → Think → Act

- **Sense:** Transaction data (amount, location, merchant type, time)
- **Think:** ML model calculates fraud probability. "Risk score = 94%"
- **Act:** Transaction declined; instant text alert sent to you

❗ **The Critical Insight :** The AI doesn't know you're on vacation. It knows the statistical pattern of "**foreign purchase followed immediately by high-value electronics**" correlates strongly with stolen card data. Pattern recognition - not understanding is what makes it work.



AI in Education - The AI Tutor

An AI tutor acts like a personal coach that watches every step of your work. Instead of just marking answers right or wrong, it figures out *why* you made a mistake and helps you fix it yourself.

What AI Does: Adapts learning content, provides instant feedback, assists with research.

AI in Education - The AI Tutor



You Type Steps

AI reads every line you type and spots errors instantly.

Example : Khanmigo (Khan Academy)



You Photograph Homework

AI scans your handwriting and explains exactly where you went wrong.

Example : Google Socratic



You Work on Screen

AI watches every click, drag, and tap you make in real time.

Example : DreamBox

AI in Entertainment

Filmmakers use AI to create visual effects that would be impossible or incredibly time-consuming to do by hand. One powerful use is **de-aging** - making an actor appear decades younger for flashback scenes.

What AI Does: Recommends content, generates special effects, personalizes experiences.

- ❶ **Example:** In *Indiana Jones and the Dial of Destiny*, ILM used machine learning to scour decades of Lucasfilm archives - including unused footage from the original trilogy - to create a digital "mask" of a young **Harrison Ford**, mapping his 1944 appearance onto his 80-year-old performance without replacing the actor's physical presence.



AI in Entertainment

SENSE

Camera captures the actor's current performance - every facial movement and expression.

THINK

AI model (trained on younger photos of the same actor) identifies facial landmarks and generates a de-aged version of each frame, matching skin texture, lighting, and expression.

ACT

The de-aged footage is rendered and integrated into the final film.

- ✓ **Human-in-the-Loop:** The AI generates a mathematically plausible younger face. But a team of human artists reviews every frame to ensure the performance still feels emotionally authentic. The subtle fear, joy, or sadness in the actor's eyes must be preserved - that's something a human can judge.



AI in Transportation

Modern aeroplanes, trains, and even delivery trucks are filled with sensors that constantly monitor engines, brakes, and other critical parts. AI analyses this stream of data in real time - this is called **predictive maintenance**.

What AI Does: Optimizes routes, predicts maintenance, powers autonomous features.

AI in Transportation

Example: Airline Predictive Maintenance

Sensors on a jet engine continuously stream vibration and temperature data. The AI detects a subtle frequency shift - a signature that, in historical data, appeared **three weeks before a specific bearing failed.**

Sense → Think → Act

- **Sense:** Engine sensors capture high-frequency vibration data
- **Think:** ML model compares real-time signature to known failure precursors. Predicts: "Bearing #4 has 85% probability of failure within 30 flight cycles."
- **Act:** Maintenance crew alerted to replace bearing during next overnight stop

Mini Activity 2: Who's Using What AI?

For each real-world AI application below, identify: the industry, the primary AI domain, and one reason a human must stay in the loop.

Drone Crop Map

A drone flies over a cornfield and creates a "stress map" showing exactly which 10-foot sections need more water or fertilizer.

2AM Purchase Alert

A bank's system flags a \$500 online purchase made at 2:00 AM from a device never used before and sends a text alert.

Reading Coach App

An app listens to a student read aloud and highlights words they mispronounced, then suggests practice for those specific sounds.

The Playlist

A music streaming app notices you've been playing "Lo-fi Beats" every rainy Tuesday, so it builds a custom "Rainy Day" playlist just for you.

Food Recommendation

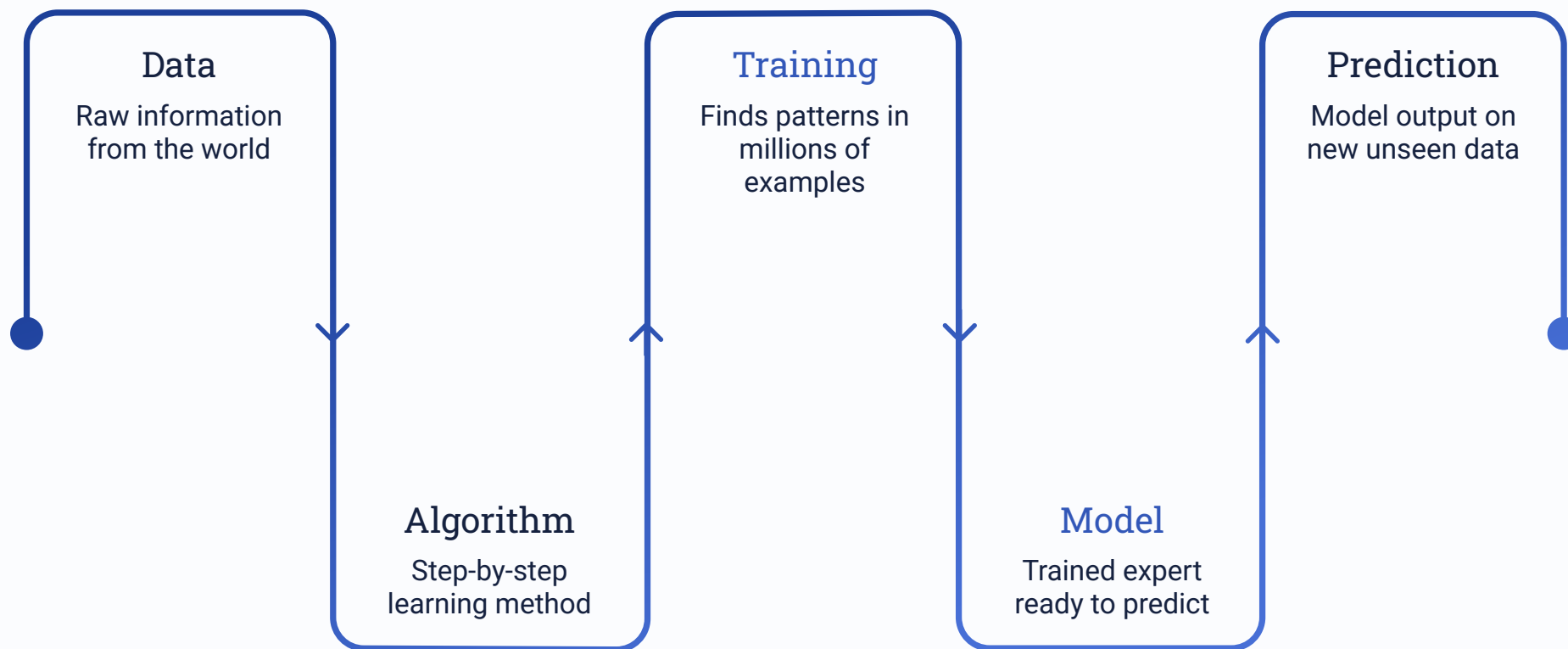
A fast-food kiosk suggests adding a chocolate milkshake to your order because the weather is over 30°C and most people today are buying cold drinks.

Mini Activity 2 - Answer Key

Application	Industry	AI Domain	Why Human Must Stay in the Loop
Drone crop stress map	Agriculture	Computer Vision (Multispectral)	AI identifies stressed plants but doesn't know why—drought, disease, or pest. A farmer must inspect and decide the right intervention.
Fraud alert on unusual purchase	Finance	Anomaly Detection (ML)	AI flags a statistical anomaly but doesn't know you're on vacation buying a gift. A human must review to avoid blocking a legitimate purchase.
Reading pronunciation coach	Education	Voice AI (ASR + Phoneme Analysis)	AI detects mispronunciation but doesn't know if the student is tired, has a speech difference, or just needs encouragement.
Personalised Music Feed	Entertainment	Recommendation Engine	AI doesn't know you're actually sad today and need upbeat pop. You can hit "skip" to tell the AI your vibe has changed.
Smart Menu Suggestions	Food & Beverage	Predictive Analytics	The AI doesn't know you have a dairy allergy. A customer must make the final choice to buy.

The AI Recipe

Across above industries and AI system you'll ever encounter - the process is identical. This is the same recipe you learned in Session 2, now applied at global scale.



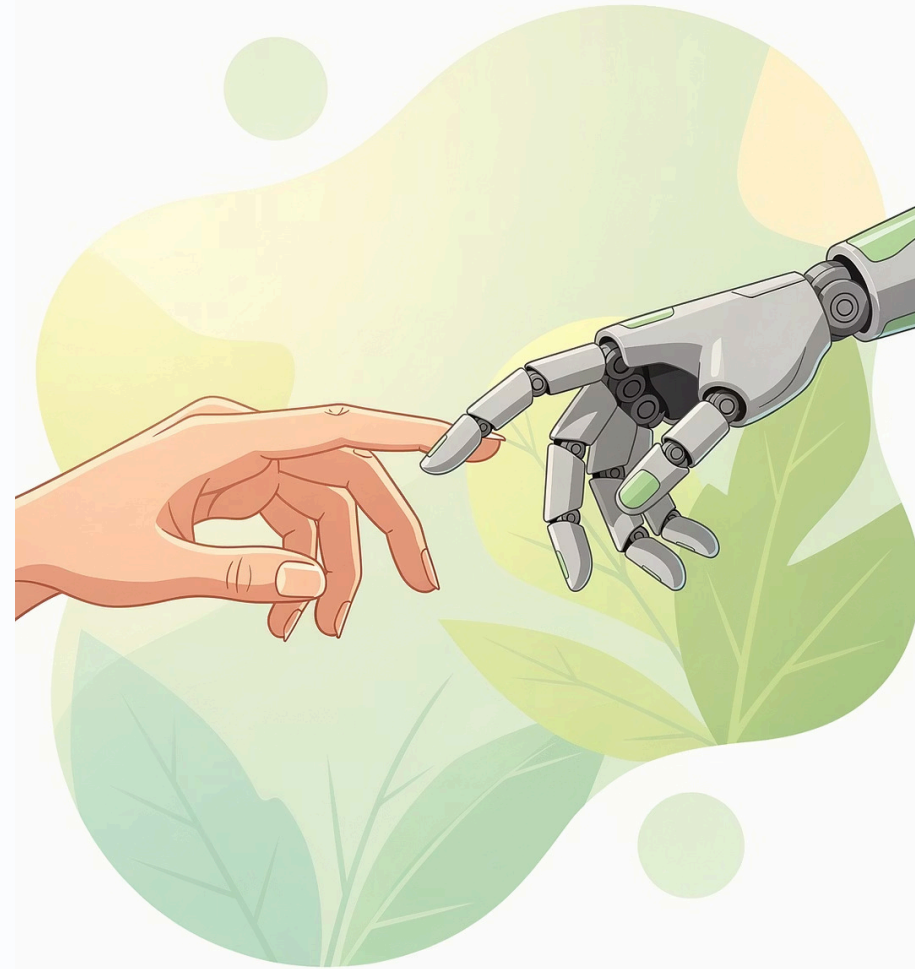
The AI Recipe

Step	What It Means	Industry Example
Data	Raw information collected from the world	X-ray images, credit card transactions, student quiz answers, game actions, jet engine vibrations
Algorithm	The step-by-step learning method	Convolutional Neural Network (images), Random Forest (fraud), Reinforcement Learning (game AI)
Training	The algorithm finds patterns by studying millions of examples	Model learns what a "nodule" looks like; learns what "fraudulent transaction pattern" means
Model	The trained expert, ready to make predictions	The finished cancer-detection AI; the fraud-detection system; the adaptive learning engine
Prediction	The model's output when shown new, unseen data	"85% probability this is a nodule"; "94% risk score this is fraud"; "Student needs practice on negative numbers"

The Human-in-the-Loop Principle

If AI is so powerful, why do we still need humans?

Because AI still lacks the one thing that defines us: Judgment.



Hands-On Activity: Industry Research

Gemini + NotebookLM

Goal: Use Gemini for broad exploration and NotebookLM for fact-verification, just like a real AI researcher would. You'll create a one-page summary on an AI industry application.

Healthcare

Cancer Detection

Finance

Fraud Prevention

Education

Personalized Learning

Entertainment

Video Game NPCs

Transportation

Predictive Maintenance

Pick one focus area above or choose an industry you want to explore

Research Activity: Step-by-Step

01

Choose Your Focus

Pick one of the industry topics: Healthcare, Finance, Education, Entertainment, Transportation or choose an industry you want to explore

02

Broad Exploration with Gemini

Open **gemini.google.com**. Ask a broad question to get an overview. Example: *"Explain how AI is used to detect lung cancer in medical scans. Keep it simple for a middle school student."* Note one interesting fact and one potential limitation.

03

Source Setup in NotebookLM

Open **notebooklm.google.com**. Create a new notebook and add the pre-vetted source links or docs.

04

Fact-Verification with NotebookLM

Ask NotebookLM the same question, but add: *"Based only on the uploaded source..."* Observe: if the source doesn't cover it, NotebookLM will say so - this is the anti-hallucination feature.

05

Compare & Create

Open a blank Google Doc. Write three bullet points: one thing Gemini taught you, one thing NotebookLM confirmed, and one reason a human expert is still essential.

Example :

Here's what a strong three-bullet AI Industry Brief looks like (Healthcare example):

Gemini Insight

Gemini explained that AI can detect lung nodules as small as **3mm** by analyzing CT scans, and mentioned that Google Health has developed a model that sometimes outperforms radiologists in spotting early cancer signs.

NotebookLM Confirmation

NotebookLM (based on the Wikipedia article) confirmed that AI models are trained on thousands of labeled scans and highlighted a challenge: models can "**overfit**" to the specific hospitals where they were trained and may not perform as well on scans from different machines.

Human Essential Role

A human radiologist is still essential because the AI only flags suspicious areas it cannot understand a patient's full medical history, symptoms, or personal context. The doctor makes the final diagnosis and decides on treatment.

Summary



Spot AI

AI is embedded in weather apps, robot vacuums, noise-canceling headphones, and more



AI Across Industries

Healthcare, Finance, Education, Entertainment, and Transportation all leverage AI for specific, high-impact tasks.



Unifying Framework

AI System follows **Sense** → **Think** → **Act**. One framework to understand them all.



Human Imperative

AI finds patterns. Humans provide judgment, ethics, and oversight. Always.



Research Skill

Use Gemini for broad ideas; use NotebookLM to verify facts from trusted sources.

The Core Message

AI + Humans

AI is a **pattern matcher**. Humans are the **judges, ethicists, and supervisors**. Across healthcare, finance, education, entertainment, transportation, etc, the most successful systems are those where AI and humans work together - AI handling the scale and speed, humans providing the wisdom and oversight.

AI Handles

Scale, speed, pattern recognition, tireless analysis

Humans Provide

Wisdom, ethics, empathy, context, and final judgment

Homework: Problem-Finder

Today you discovered that AI is everywhere. Now it's your turn to think like an AI innovator. Your mission has four short parts:

- 1 Spot the AI in Your World**
Notice 3 moments where AI touched your day. Write them down—just a phrase each.
- 2 Identify One Problem Worth Solving**
Look at your daily routine. What's one small frustration that AI could potentially help with?
- 3 Brainstorm an AI Solution**
Use Gemini or ChatGPT to quickly brainstorm how AI might help solve your problem.
- 4 Add to Your Prompt Library**
Create a new entry in your Prompt Library document with today's date and your findings.



Homework: Problem Statement Examples

Need inspiration? Here are some example problem statements to spark your thinking:

“

"I wish there were an easier way to **keep track of all my homework deadlines** in one place."

“

"I wish there were an easier way to **remember to practice guitar** every day."

“

"I wish there were an easier way to **decide what to pack for school lunch** based on what's in the fridge."

“

”

”

”

- 🕒 **Why This Homework Matters:** This assignment directly prepares you for Session 19, where you'll use Lovable to build a personalized web app that addresses a real-life problem you care about. By identifying a challenge now and using AI to brainstorm solutions, you'll arrive at the next session with a concrete idea and a sense of ownership.

Next Session Preview

Session 19: AI Careers & The Future of Work

AI Careers

Explore the careers emerging in the age of AI.

Become a Builder

Using **Lovable**, you'll design and create a simple, personalized web app that addresses a real-life problem you care about.

Your App Ideas

A homework organizer, a chore tracker, a study timer, or something entirely your own. You'll leave with a **working app**.

